

AIML SESSION 03 ASSIGNMENT SUBMISSION

Name of student: Vaibhav Vijay Kate

Domain/subject name: AIML (Artificial Intelligence & Machine Learning)

College and Branch name: Government Polytechnic College Washim (IF)

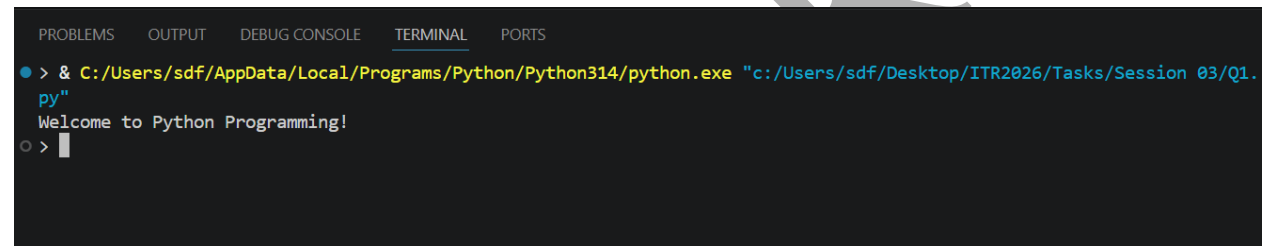
Task 1: Basic Function

Write a function named `welcome()` that prints the message: "Welcome to Python Programming!" Call the function to execute it. Add a comment explaining why functions are useful

Program Code:

```
def welcome():  
    print("Welcome to Python Programming!")  
welcome()
```

OUTPUT Screenshot:



The screenshot shows a terminal window with a dark background. At the top, there are tabs for 'PROBLEMS', 'OUTPUT', 'DEBUG CONSOLE', 'TERMINAL' (which is active), and 'PORTS'. The terminal shows a command prompt where a Python script is executed. The command is: `> & C:/Users/sdf/AppData/Local/Programs/Python/Python314/python.exe "c:/Users/sdf/Desktop/ITR2026/Tasks/Session 03/Q1.py"`. The output of the script is printed: `Welcome to Python Programming!`. Below the output, there is a prompt `o > |` with a cursor.

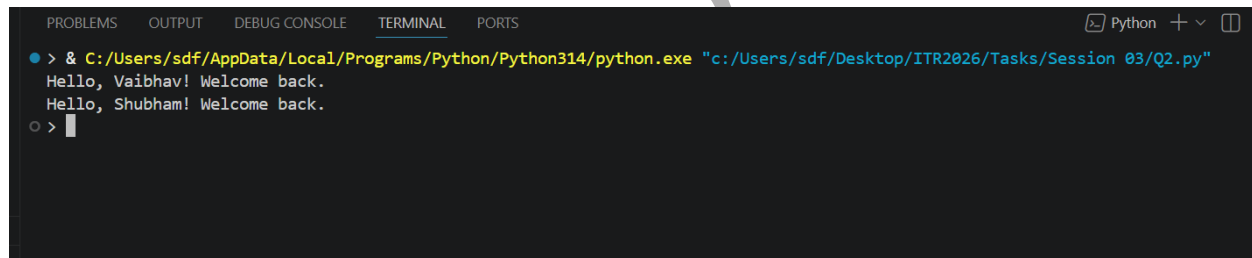
Task 2: Function with Parameter

Create a function named `greet(name)` that takes one parameter `name` and prints: "Hello, [name]! Welcome back." Call the function with your own name. Also call it with a different name to show reusability

Program Code:

```
def greet(name):  
    print("Hello,", name + "! Welcome back.")  
  
greet("Vaibhav")  
greet("Shubham")
```

OUTPUT Screenshot:



The screenshot shows a Python IDE interface with a terminal window. The terminal displays the command to run a Python script, followed by the output of the `greet` function. The output shows two lines: "Hello, Vaibhav! Welcome back." and "Hello, Shubham! Welcome back.".

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS Python + v []  
● > & C:/Users/sdf/AppData/Local/Programs/Python/Python314/python.exe "c:/Users/sdf/Desktop/ITR2026/Tasks/Session 03/Q2.py"  
Hello, Vaibhav! Welcome back.  
Hello, Shubham! Welcome back.  
○ > |
```

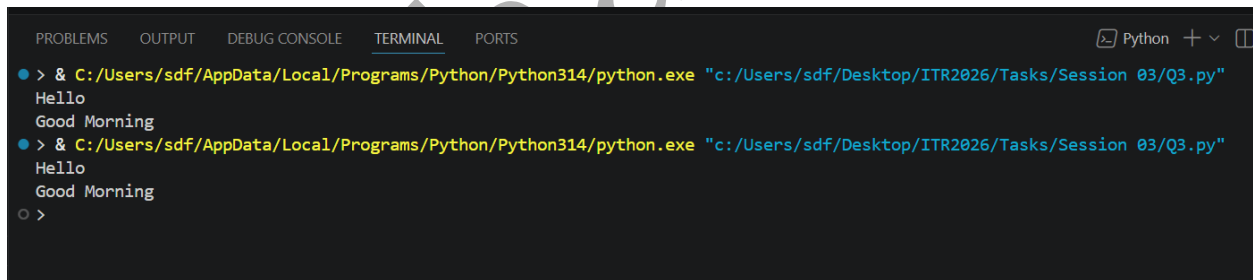
Task 3: Default Parameter

Write a function `show_message(message="Hello")` that prints the message. • Call the function without passing any argument. • Call the function again by passing a different message. Explain the benefit of default parameters in a comment.

Program Code:

```
def show_message(message="Hello"):  
    print(message)  
  
show_message()  
show_message("Good Morning")
```

OUTPUT Screenshot:



```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  Python + -
```

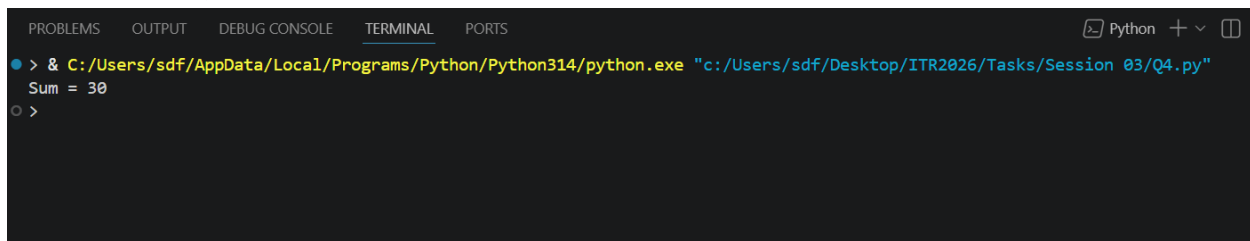
```
> & C:/Users/sdf/AppData/Local/Programs/Python/Python314/python.exe "c:/Users/sdf/Desktop/ITR2026/Tasks/Session 03/Q3.py"  
Hello  
Good Morning  
> & C:/Users/sdf/AppData/Local/Programs/Python/Python314/python.exe "c:/Users/sdf/Desktop/ITR2026/Tasks/Session 03/Q3.py"  
Hello  
Good Morning  
o >
```

Task 4: Function with Multiple Parameters & Return) Create a function `calculate_sum(a, b)` that takes two numbers and returns their sum. • Call the function with two numbers and store the result in a variable. • Print the result. Also demonstrate the difference between `print()` inside the function and using `return`.

Program Code:

```
def calculate_sum(a, b):  
    return a + b  
  
result = calculate_sum(10, 20)  
print("Sum =", result)  
  
def add_print(a, b):  
    print("Sum inside function =", a + b)  
  
def add_return(a, b):  
    return a + b  
add_print(5, 3)  
ans = add_return(5, 3)  
print("Returned Sum =", ans)
```

OUTPUT Screenshot:



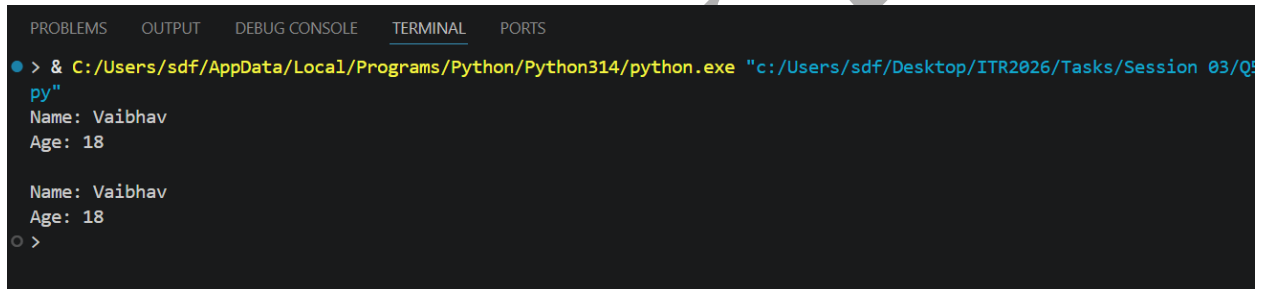
```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  
● > & C:/Users/sdf/AppData/Local/Programs/Python/Python314/python.exe "c:/Users/sdf/Desktop/ITR2026/Tasks/Session 03/Q4.py"  
Sum = 30  
○ >
```

Task 5: Positional vs Keyword Arguments) Write a function `student_info(name, age)` that prints the name and age. Call this function twice

Program Code:

```
def student_info(name, age):  
    print("Name:", name)  
    print("Age:", age)  
student_info("Vaibhav", 18)  
print()  
student_info(age=18, name="Vaibhav")
```

OUTPUT Screenshot:



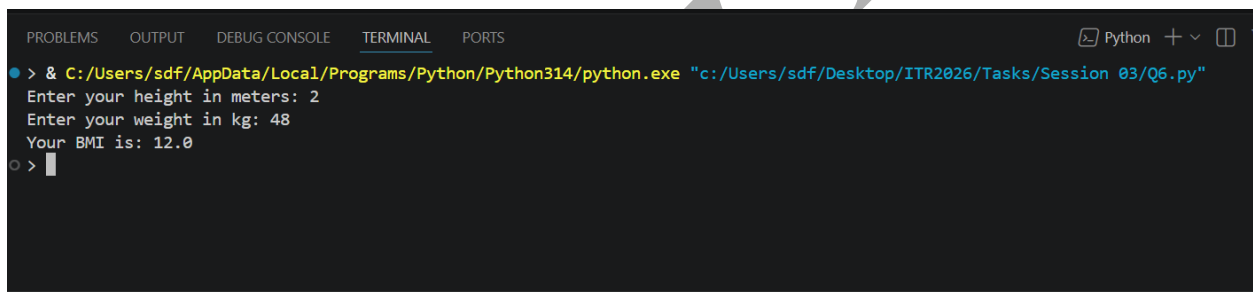
```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS  
● > & C:/Users/sdf/AppData/Local/Programs/Python/Python314/python.exe "c:/Users/sdf/Desktop/ITR2026/Tasks/Session 03/Q5.py"  
Name: Vaibhav  
Age: 18  
  
Name: Vaibhav  
Age: 18  
○ >
```

Task 6: Function with Return Value) Write a function `square(num)` that returns the square of the given number. Write another function `cube(num)` that returns the cube of the number. In the main program, take a number as input from the user and print its square and cube using these functions

Program Code:

```
height = float(input("Enter your height in meters: "))
weight = float(input("Enter your weight in kg: "))
bmi = weight / (height ** 2)
print("Your BMI is:", round(bmi, 2))
```

OUTPUT Screenshot:



The screenshot shows a terminal window with the following content:

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS Python + - []
> & C:/Users/sdf/AppData/Local/Programs/Python/Python314/python.exe "c:/Users/sdf/Desktop/ITR2026/Tasks/Session 03/Q6.py"
Enter your height in meters: 2
Enter your weight in kg: 48
Your BMI is: 12.0
>
```

Task 7: Recursion - Basic) Write a recursive function `countdown(n)` that prints numbers from `n` down to 1. Call `countdown(10)`. Make sure to include the base case (stopping condition).

Program Code:

```
"""This program calculates the area and perimeter of a rectangle using user-provided dimensions."""
```

```
length = float(input("Enter length: "))
```

```
width = float(input("Enter width: "))
```

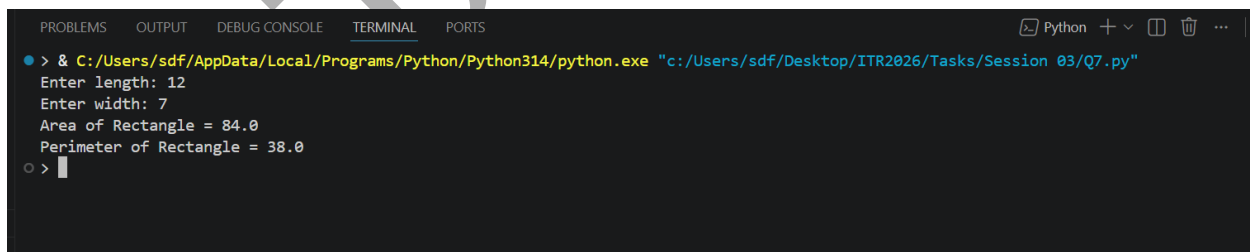
```
area = length * width
```

```
perimeter = 2 * (length + width)
```

```
print("Area of Rectangle =", area)
```

```
print("Perimeter of Rectangle =", perimeter)
```

OUTPUT Screenshot:



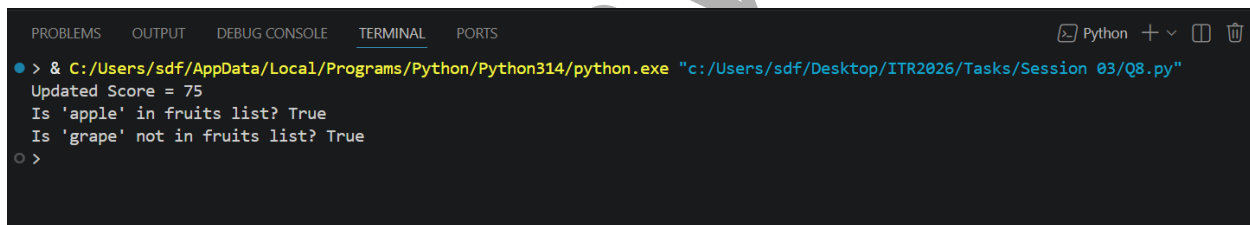
```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS Python + - [ ] [ ] ... |
> & C:/Users/sdf/AppData/Local/Programs/Python/Python314/python.exe "c:/Users/sdf/Desktop/ITR2026/Tasks/Session 03/Q7.py"
Enter length: 12
Enter width: 7
Area of Rectangle = 84.0
Perimeter of Rectangle = 38.0
> |
```


Task 8: Recursion - Factorial) Write a recursive function `factorial(n)` that calculates and returns the factorial of a number. Example: `factorial(5)` should return 120. Test it with input from the user and print the result. Add a comment explaining the base case.

Program Code:

```
fruits = ["mango", "apple", "banana", "cherry"]
score = 50
score += 25
print("Updated Score =", score)
print("Is 'apple' in fruits list?", "apple" in fruits)
print("Is 'grape' not in fruits list?", "grape" not in fruits)
```

OUTPUT Screenshot:

A screenshot of a code editor's terminal window. The terminal has tabs for PROBLEMS, OUTPUT, DEBUG CONSOLE, TERMINAL (selected), and PORTS. The command prompt shows the execution of a Python script: `> & C:/Users/sdf/AppData/Local/Programs/Python/Python314/python.exe "c:/Users/sdf/Desktop/ITR2026/Tasks/Session 03/Q8.py"`. The output displayed is: `Updated Score = 75`, `Is 'apple' in fruits list? True`, and `Is 'grape' not in fruits list? True`. The prompt `o >` is visible at the bottom.

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS Python + - [ ] [X]
> & C:/Users/sdf/AppData/Local/Programs/Python/Python314/python.exe "c:/Users/sdf/Desktop/ITR2026/Tasks/Session 03/Q8.py"
Updated Score = 75
Is 'apple' in fruits list? True
Is 'grape' not in fruits list? True
o >
```

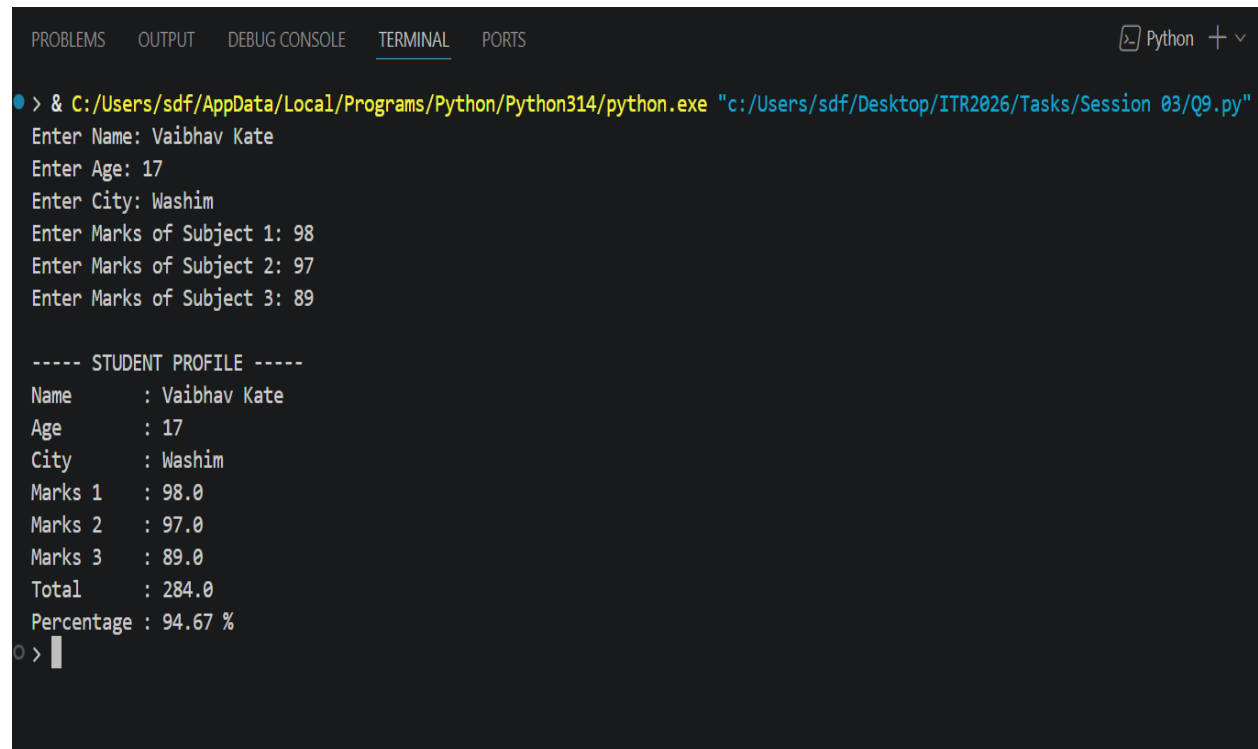
Task 9: Scope - Local vs Global) Write a program to demonstrate scope:

- Create a global variable `total = 0`
- Write a function `add_value(x)` that adds `x` to `total` (use the `global` keyword). Call the function multiple times and print the value of `total` after each call. Also try to print a local variable outside its function to show it doesn't work.

Program Code:

```
name = input("Enter Name: ")
age = int(input("Enter Age: "))
city = input("Enter City: ")
mark1 = float(input("Enter Marks of Subject 1: "))
mark2 = float(input("Enter Marks of Subject 2: "))
mark3 = float(input("Enter Marks of Subject 3: "))
total_marks = mark1 + mark2 + mark3
percentage = total_marks / 3
print("\n----- STUDENT PROFILE -----")
print("Name    :", name)
print("Age     :", age)
print("City    :", city)
print("Marks 1   :", mark1)
print("Marks 2   :", mark2)
print("Marks 3   :", mark3)
print("Total    :", total_marks)
print("Percentage :", round((percentage, 2), "%"))
```

OUTPUT Screenshot:



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS Python + v
● > & C:/Users/sdf/AppData/Local/Programs/Python/Python314/python.exe "c:/Users/sdf/Desktop/ITR2026/Tasks/Session 03/Q9.py"
Enter Name: Vaibhav Kate
Enter Age: 17
Enter City: Washim
Enter Marks of Subject 1: 98
Enter Marks of Subject 2: 97
Enter Marks of Subject 3: 89

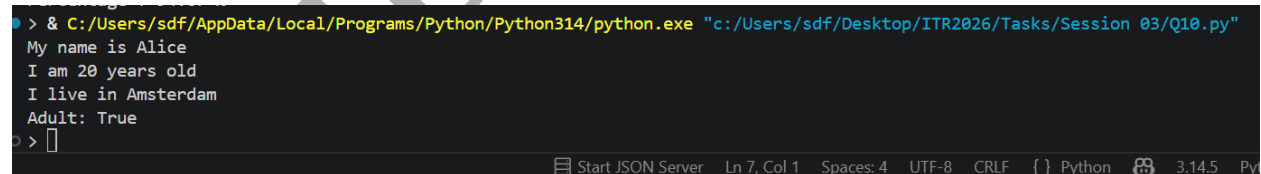
---- STUDENT PROFILE ----
Name      : Vaibhav Kate
Age       : 17
City      : Washim
Marks 1   : 98.0
Marks 2   : 97.0
Marks 3   : 89.0
Total     : 284.0
Percentage : 94.67 %
○ > |
```

Task 10: Mini Project - All Concepts) Create a Simple Number Analyzer program using functions: Write the following functions: get_number() → takes input from user and returns the number. Assignment Questions is_even(num) → returns True if number is even, else False. power(base, exp=2 → returns base raised to exponent (use default argument). show_result(num) → prints whether the number is even/odd and its square. Use a while loop to repeatedly ask the user for numbers until they enter 0. Use recursion only if needed for one small part.

Program Code:

```
name = "Alice"
age = 20
city = "Amsterdam"
print("My name is", name)
print("I am", age, "years old")
print("I live in", city)
print("Adult:", age > 18)
```

OUTPUT Screenshot:



```
> & C:/Users/sdf/AppData/Local/Programs/Python/Python314/python.exe "c:/Users/sdf/Desktop/ITR2026/Tasks/Session 03/Q10.py"
My name is Alice
I am 20 years old
I live in Amsterdam
Adult: True
> 
```